

DATA SPLIT

$G, X, Y_{\text{train}}, Y_{\text{val}}$

Fixed split



$G = (V, E), X$

train val test

EXPERT TRAINING

same split, same budget

identical data split and optimization budget

Expert H: self-loop residual

self-loop graph



filter bank



node gate



HARP-GNN

a_H
val score

Expert E: ego-separated no-self

ego view



no-self filter

$$B_k = \tilde{A}^k B_0$$

residual fusion

$[B_0, U]$

HARP-ESep

a_E
val score

ROUTING

validation labels only

route first, evaluate second

Confidence router

$$\Delta_{\text{val}} = a_E - a_H$$



1.96SE
switch only with
clear validation evidence

freeze before test

TEST

after branch lock

Locked expert



test score
+ oracle regret

test labels
never route